Boyu Yao

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RESEARCH INTERESTS

Power system planning, electricity markets, and resource adequacy under physical, operational, market, and policy constraints; critical minerals, energy supply chains, and grid infrastructure; emerging technologies and large-scale data center electricity demand.

EDUCATION

Johns Hopkins University **Aug. 2023 – Present**

Ph.D. in Civil and Systems Engineering
Advisor: Prof. Yury Dvorkin
Affiliations: Ralph O'Connor Sustainable Energy Institute (ROSEI); Critical Minerals Research Lab; Electric Power Innovation for a Carbon-free Society (EPICS); Center for Systems Science and Engineering (CSSE); Academic Center for Reliability and Resilience of Offshore Wind (ARROW)

Johns Hopkins University **Jan. 2021 – May 2022**

Master of Science in Engineering, Geography and Environmental Engineering
Advisor: Prof. Ben Hobbs
Concentration: Environmental Management and Economics

Tianjin University **Sept. 2016 – June 2020**

Bachelor of Engineering, Environmental Engineering
3+2 Program with **University of California, Irvine**, Henry Samueli School of Engineering
Exchange Student in Civil and Environmental Engineering Aug. 2019 – June 2020

PUBLICATIONS & RESEARCH OUTPUTS

Preprints & Manuscripts Under Review

- Yuhan Dai, **Boyu Yao**: *Massachusetts' 2026 Clean Peak Standard Recalibration: Adaptation and Storage Trade-offs.* Accepted by the *17th IEEE International Conference on Green Energy & Smart Systems (IEEE GESS 2026)*, Long Beach, CA, Nov. 2026; *arXiv preprint*, Sep. 2026.
- Saroj Khanal, Geon Roh, **Boyu Yao**, Abraham Silverman, Dennice Gayme, Charalambos Konstantinou, Jip Kim, Yury Dvorkin: *Shift or Curtail? How Much Data-Center Flexibility Is Worth Depends on the Host Power Grid.* Under revision at *Nature Energy*; *arXiv preprint*, Aug. 2026.
- **Boyu Yao**, Yury Dvorkin: *Grid-Supporting Equipment Supply Chains Constrain the Feasible Pace of Power System Expansion.* Accepted at *Cell Reports Sustainability*; *arXiv preprint*, Apr. 2026.

Peer-Reviewed Articles

- Owen Reed, **Boyu Yao**, Saroj Khanal, Magdalena Klemun, Yury Dvorkin: *Electricity Prices under Rapid Load Growth: The Case of Data Centers in Virginia. Oxford Energy Forum*, Mar. 2026.
- Ming Wei, **Boyu Yao**, Michael Davidson: *Resource Adequacy Under Institutional Constraints and the Low-Carbon Energy Transition in China. Energy Policy*, Mar. 2026.
- **Boyu Yao**, Andrey Bernstein, Yury Dvorkin: *Generation Expansion Planning with Upstream Supply Chain Constraints on Materials, Manufacturing, and Deployment. IEEE Transactions on Energy Markets, Policy, and Regulation*, Jan. 2026.
- **Boyu Yao**, Hannah Jeong, Mahdi Mehrtash, Bentley Allan, Daniel Ockerman, Yury Dvorkin: *Understanding Supply Chain Constraints for the U.S. Clean Energy Transition. Nature Portfolio Journal Clean Energy*, Sep. 2025.

- **Boyu Yao**, Yury Dvorkin: *A Decision Framework for Clean Energy Supply Chain Optimization.* In *Proceedings of the 2025 IEEE Power & Energy Society General Meeting (PESGM)*, IEEE PESGM, Austin, Texas, USA, July 2025.

Technical Reports, Policy Briefs & Datasets

- Saroj Khanal, **Boyu Yao**, Abraham Silverman, Yury Dvorkin: *Valuing Data Center Load Shifting and Curtailment in PJM.* Policy brief, Ralph O'Connor Sustainable Energy Institute (ROSEI), Johns Hopkins University, ROSEI Policy Brief, Sep. 2026.
- Saroj Khanal, **Boyu Yao**, Marisel Villafañe-Delgado, Yury Dvorkin: *ICARUS PJM Dataset (ICARUS-PJM).* Versioned dataset, GitHub, Sep. 2026.
- **Boyu Yao**, Abe Silverman, Yury Dvorkin: *The Next Data Center Bottleneck Is Grid-Supporting Equipment.* Policy brief, Ralph O'Connor Sustainable Energy Institute (ROSEI), Johns Hopkins University, ROSEI Policy Brief, Apr. 2026.
- Owen Reed, Saroj Khanal, **Boyu Yao**, Dyer Combariza, Magdalena Klemun, Yury Dvorkin: *ICARUS Data Center Dataset (ICARUS-DC).* Versioned dataset, GitHub, May 2026.
- Yury Dvorkin, Benjamin Hobbs, Ziting Huang, Stephanie Wilcox, **Boyu Yao**: *Energy Resilience and Efficiency in Maryland.* Technical report for the Energy Resilience and Efficiency Working Group, State of Maryland, Maryland EREWG, Dec. 2024.
- Ming Wei, **Boyu Yao**, Chi Gao, Max Dupuy, Michael Davidson: *North China and Northeast China Interprovincial Electricity Trade Brief.* Electricity Market Tracker, Oct. 2023.
- Fikri Kucuksayacigil, **Boyu Yao**, Ming Wei, Chi Gao, Max Dupuy, Michael Davidson: *Northeast China Grid Winter Resource Adequacy Report.* Electricity Market Tracker, July 2023.

RESEARCH & PROFESSIONAL APPOINTMENTS

Resources for the Future **Washington, DC & Remote**

Research Scholar, Critical Minerals Research Lab Sep. 2026 – Jul. 2027

- Participate in the 2026–2027 research cohort, provide feedback on fellow scholars' research, and engage in policy discussions and collaborative research on critical minerals.

Researcher, RFF Subaward Project Sep. 2026 – Jan. 2027

Supervisors: Karen Palmer and Jenya Kahn-Lang

- Continued research initiated during the summer internship through an RFF-funded subaward to Johns Hopkins University, supported by the Blue Horizons initiative, focusing on resource adequacy, forward contracting, and strategic market behavior.

Research Intern Jun. 2026 – Aug. 2026

Supervisors: Karen Palmer and Jenya Kahn-Lang

- Conducted research on capacity markets, resource adequacy, and electricity market design by developing and evaluating analytical models and decision-support tools.

Johns Hopkins University **Baltimore, MD**

Graduate Research Assistant Aug. 2023 – Present

Advisor: Prof. Yury Dvorkin, Smart Energy reseARCH (SEARCH) Group

- **Clean Energy Supply Chains:** Designed a national framework linking critical material supply chains to U.S. decarbonization targets and clean-energy deployment feasibility.
- **Supply-Chain-Constrained Capacity Expansion:** Built an optimization model to quantify how critical material, manufacturing, and land constraints shape generation capacity expansion.
- **Grid-Supporting Equipment Supply Chains:** Integrated stock-flow, bill-of-materials, supply-use, and expansion modeling to assess how equipment and material constraints shape power system expansion.
- **Maryland Energy Resilience and Efficiency:** As part of the Maryland Energy Resilience and Efficiency Working Group (EREWG), analyzed electricity storage pathways and grid transformation strategies to enhance resilience and support Maryland's 2031 decarbonization targets.

- **Advanced Nuclear Integration to PJM:** In collaboration with Constellation, assessed Gen IV nuclear flexibility and PJM market impacts under rising data center demand using capacity expansion modeling.

Research Assistant Jan. 2022 – May 2022
Advisor: Prof. Ben Hobbs

- Enhanced the JHSMINE energy optimization model by adding variables and constraints, updating system cost calculations, and automating data processing and visualization.

University of California, San Diego **San Diego, CA**

Research Associate Aug. 2022 – Aug. 2023
Supervisor: Prof. Michael Davidson, Power Transformation Lab

- **Winter Resource Adequacy Analysis:** Conducted power system reliability analyses for the Northeast China Grid, providing insights into resource adequacy, seasonal demand, and renewable energy integration.
- **Interprovincial Electricity Trade Analysis:** Assessed the impact of long-term contracts on system costs and reliability in China's power grid, proposing alternative designs through simulation-based analysis.

Tsintergy Technology **Guangzhou, Guangdong, China**

Consultant May 2021 – Dec. 2021
Energy Operations Research Center, Tsinghua University

- Analyzed Guangdong's energy generation and market mechanisms and developed a carbon emissions calculation guide based on IPCC methodology.
- Supported renewable energy policy analysis and energy demand forecasting for Qinghai and Jiangsu through strategy assessment, MATLAB model enhancement, grey modeling, and emissions estimation.

TEACHING & MENTORING EXPERIENCE

Teaching Experience

- **Teaching Assistant, Energy Systems** — Johns Hopkins University, Fall 2024, Fall 2025, Fall 2026. Instructor: Prof. Yury Dvorkin. Led weekly office hours and graded assignments and exams for 30+ students; supported instruction in unit commitment, economic dispatch, and optimal power flow.
- **Teaching Assistant, Energy Policy and Planning Models** — Johns Hopkins University, Spring 2022. Instructor: Prof. Ben Hobbs. Graded assignments and exams for 20+ students and supported review sessions on energy policy and power system planning models.

Research Mentorship

- **Luca Montoya** — JHU ROSETAS REU, Summer 2025. Resource adequacy assessment for capacity expansion planning.
- **Princeton Nguyen** — JHU ROSETAS REU, Summer 2025. Weather-informed scenario generation for capacity expansion planning in ISO-NE.
- **Maria Vabson** — Johns Hopkins University, Sep. 2023 – May 2024. Critical material identification, stock assessment, and production metrics for clean-energy supply chains.

SELECTED PRESENTATIONS

Invited Talks

- *Massachusetts' 2026 Clean Peak Standard Recalibration: Adaptation and Storage Tradeoffs.* The 17th IEEE International Conference on Green Energy and Smart Systems, 2026.
- *AI Data Centers Need More Than Power: Transformers and Power Electronics Are Hidden Supply Chain Externalities.* INFORMS Annual Meeting, 2026.
- *Critical Minerals as Constraints on Power System Expansion.* Critical Minerals Research Lab Research Scholar Conference, 2026.
- *Beyond Market Signals: Supply-Chain Constraints in Grid Expansion.* Power Markets Journal Club, 2026.

- *Market Signals and Incentives Aren't Sufficient: The Case for Supply-Chain Modeling for Grid Expansion.* INFORMS Annual Meeting, 2025.
- *A Fully Open-Source Power Dataset for Policy Applications in China.* INFORMS Annual Meeting, 2023.

Oral Presentations

- *When the Supply Chain Shapes the Grid: Integrating Upstream Constraints into Long-Term Planning.* JHU Energy and Environment Systems Seminar, 2025.
- *Integrating Upstream Supply Chains into Generation Expansion Planning.* JHU Center for Systems Science and Engineering (CSSE) Meeting, 2025.
- *Energy Resilience and Efficiency Analyses to Support the Maryland Climate Change Commission.* Annual Meeting of the Energy Resilience and Efficiency Working Group, Maryland Department of the Environment, 2024.
- *Power System Resource Adequacy in the Low-Carbon Transition—Northeast China Grid Winter Resource Adequacy.* 30th Technical Workshop of the China Energy Modeling Forum (CEMF), 2023.

Poster Presentations

- *Grid-Supporting Equipment Supply Chains as a Constraint on Power System Expansion.* PowerUp Conference, 2026.
- *The ICARUS Dataset: An Infrastructure-Centric, Open Dataset for Capacity Expansion, Resource Adequacy, and Data Center Integration.* Ralph O'Connor Sustainable Energy Institute (ROSEI) Summit, 2026.
- *When the Supply Chain Shapes Long-Term Grid Planning.* Ralph O'Connor Sustainable Energy Institute (ROSEI) Summit, 2026.
- *When the Supply Chain Shapes Long-Term Grid Planning.* Electric Power Innovation for a Carbon-free Society (EPICS) Center U.S. Summit, 2025.
- *A Decision Framework for Clean-Energy Supply-Chain Optimization.* IEEE Power & Energy Society General Meeting, 2025.
- *Understanding Supply Chain Constraints for the U.S. Energy Transition.* JHU Sustainability Symposium on Research, Practice, and Education, 2025.
- *Understanding Clean-Energy Supply Chains: A U.S. Perspective.* Ralph O'Connor Sustainable Energy Institute (ROSEI) Summit, 2025.

SERVICE

- **Peer Reviewer** — Ongoing. Reviewed manuscripts for *IEEE Transactions on Energy Markets, Policy and Regulation*, *IEEE Access*, *Mitigation and Adaptation Strategies for Global Change*, *BMC Environmental Science*, etc.
- **Graduate Orientation Mentor** — Johns Hopkins University, Civil and Systems Engineering, 2024 – Present. Mentored incoming graduate students during departmental orientation and their transition into the program.
- **Volunteer Science Teacher** — Chiang Mai, Thailand, Oct. 2017. Developed and taught science lessons for primary school students.

AWARDS & HONORS

- **RFF Blue Horizons Research Subaward** (*Resources for the Future; Johns Hopkins University*) **2026–2027**
- **IEEE PES General Meeting Travel Fellowship** (*IEEE Power & Energy Society*) **2025**
- **Joseph Meyerhoff Fellowship** (*Johns Hopkins University*) **2023**
- **The 20th "Students of Science & Technology Outstanding Talents"** (*Tianjin University*) **2019**
- **Youth "Eco-Innovation Green Action" Competition** (*National, Third Prize*) **2019**
- **The 15th "Challenge Cup" National Undergraduate Curricular Academic Science & Technology Competition** (*Provincial, First Prize*) **2019**
- **The 12th "Challenge Cup" National Undergraduate Curricular Academic Science & Technology Competition** (*Tianjin University, First Prize*) **2019**
- **The 4th Tianjin College Students Environmental Discipline Innovation & Practice Invitational Tournament** (*Provincial, Second Prize*) **2018**

- **Representative of 3rd Plenary of the Fifth Council, Chinese Society for Engineering Education (CSEE)** (*"Integration of New Engineering Construction with Industry and Education" Int'l Conference*) **2018**
- **Merit Student** (*Tianjin University*) **2017**